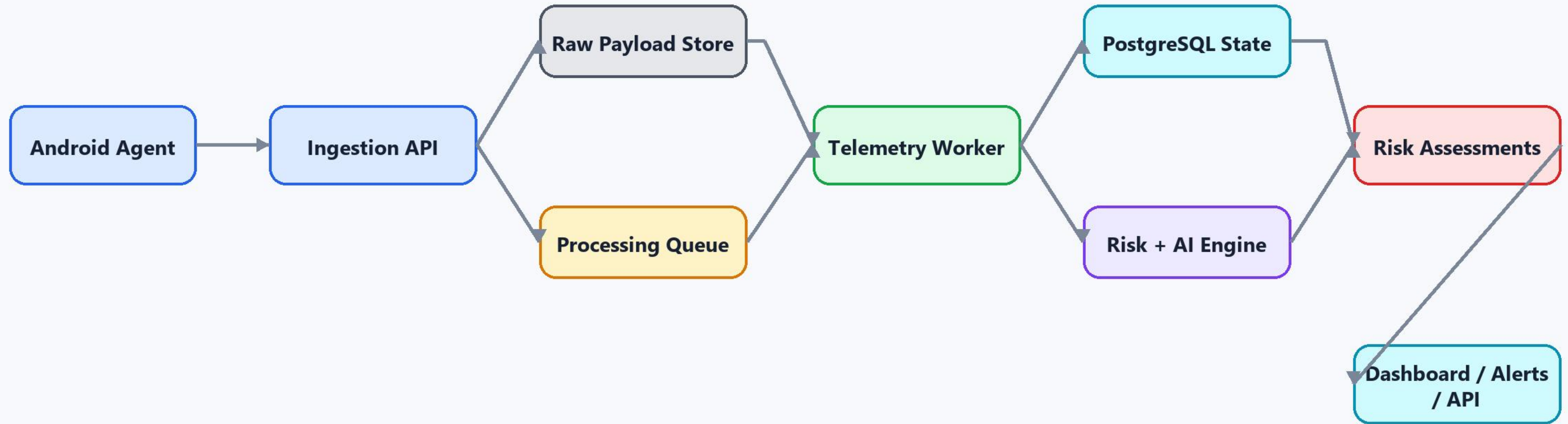


AEGIS Final Backend + AI Architecture

Simple production backend with AI used only where it adds security value



Ingestion stays simple

- auth
- schema validation
- idempotency
- raw storage

Worker does processing

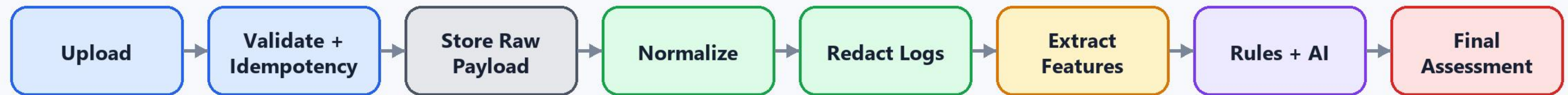
- normalize telemetry
- redact logs
- extract features
- update state

AI is controlled

- rules first
- LLM only if needed
- evidence required
- auditable outputs

Final Processing Flow

From agent upload to final risk assessment



Reliability

- same payload_id is safe to retry
- raw payload is kept for replay
- worker can reprocess later

Data engineering

- posture/app/log normalization
- latest app state
- compact feature vector

AI result

- score
- reasons
- confidence
- recommended action

Simple Risk + AI Decision Flow

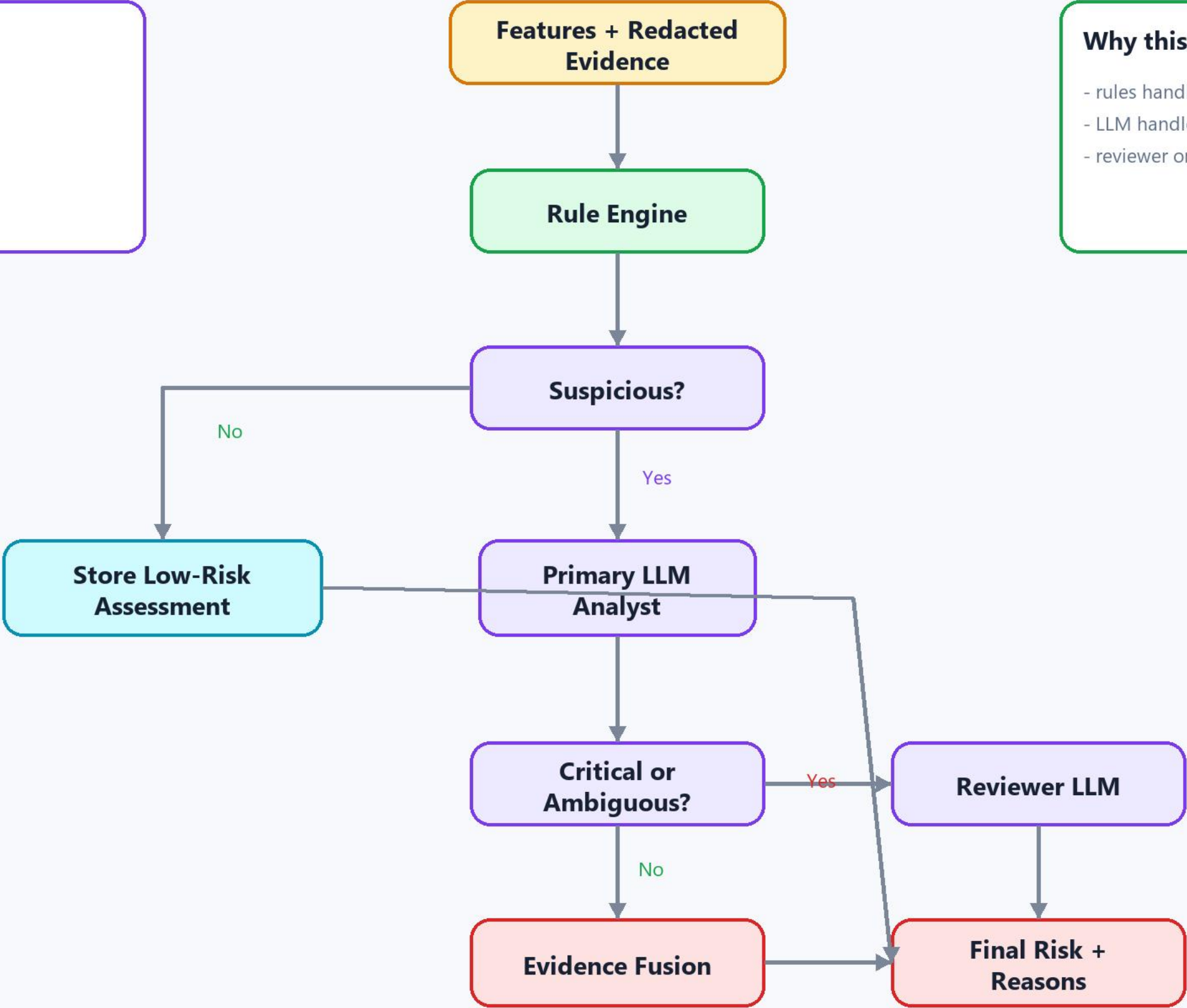
Rules always run; LLMs run only for suspicious or ambiguous cases

LLM guardrails

- redacted input only
- structured JSON output
- must cite evidence
- cannot execute actions

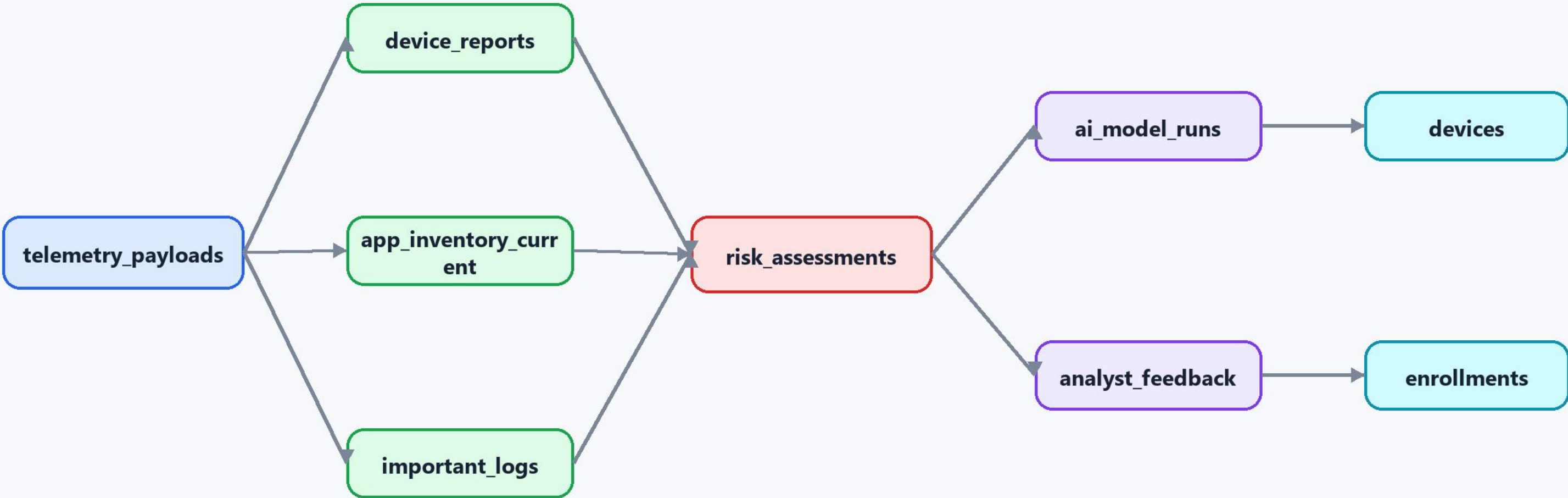
Why this is simple

- rules handle normal cases
- LLM handles only useful cases
- reviewer only for critical cases



Minimal Backend Data Model

Small first schema that still supports replay, scoring, AI audit, and feedback



Source of truth

- telemetry_payloads owns payload_id idempotency
- raw payload URI supports replay

Security state

- device posture
- latest app inventory
- redacted important logs

Decision layer

- risk score and reasons
- model audit trail
- analyst feedback